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GOSIM Shenzhen 2026

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How Do You Read a Million Tokens on Your Laptop? You Don't.

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01

About Pleias



We train small open language models for working with documents

Open data

Trained on Common Corpus, a large open dataset of public-domain and openly licensed text

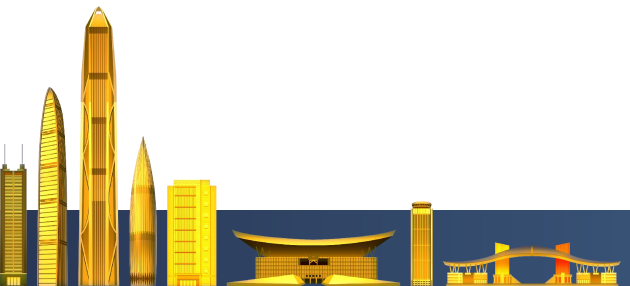
Small models

Our reasoning model Baguettotron has only 321M parameters

Built for documents

Answers stay grounded in the sources and cite them

Open weights on Hugging Face: huggingface.co/PleIAs

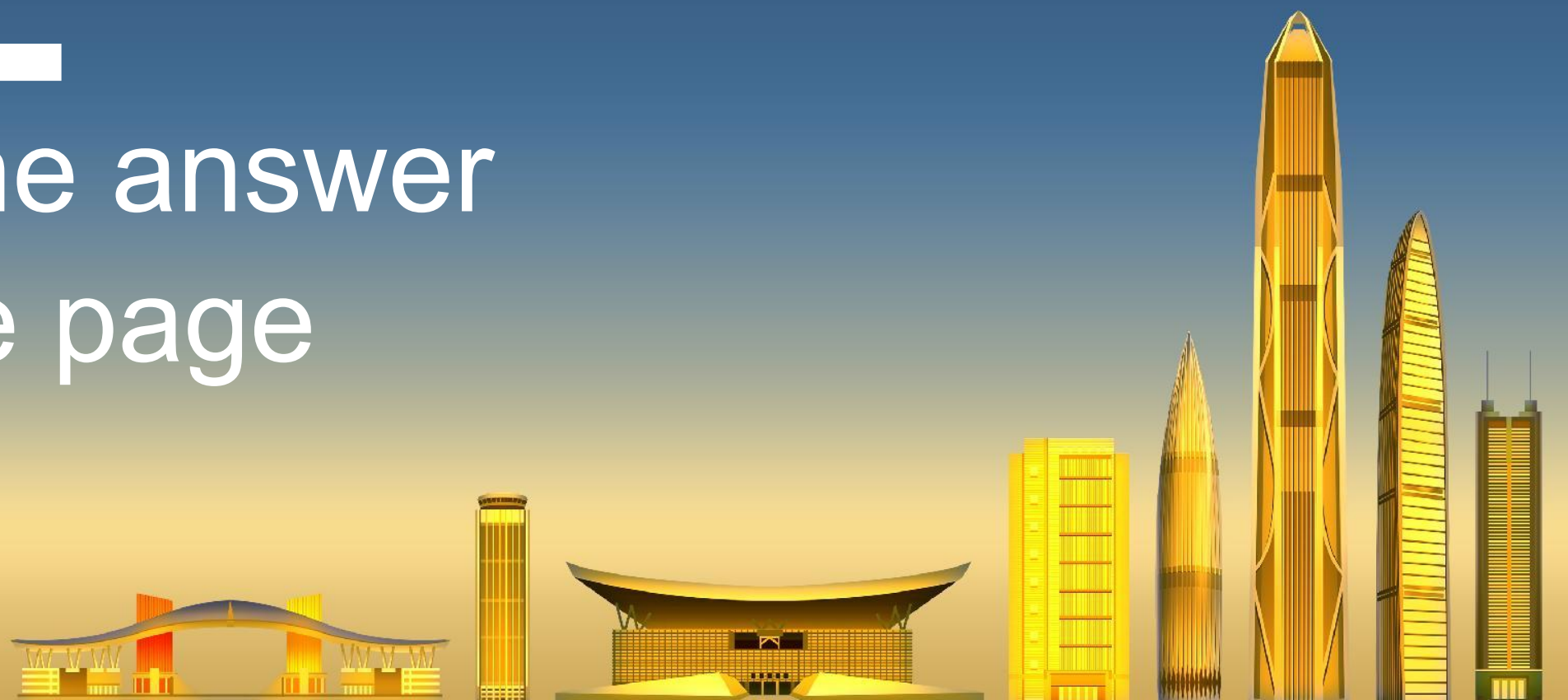


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02

When the answer is on the page



An offline legal assistant that answers with exact quotes from its source, on a Raspberry Pi 5

Question

Our government is negotiating a peace deal. Can it include an amnesty for soldiers who raped civilians during the war?

Answer (model output, trimmed)

Amnesty provisions are generally not considered permissible in cases of Conflict-Related Sexual Violence (CRSV), particularly when it involves serious international crimes. [1] *“…exclude the application of amnesty provisions.”*

[1] Red Line Initiative Guidebook › UN Peace & Security › III.12

328 MB

321M Baguettotron fine-tune

28 tok/s

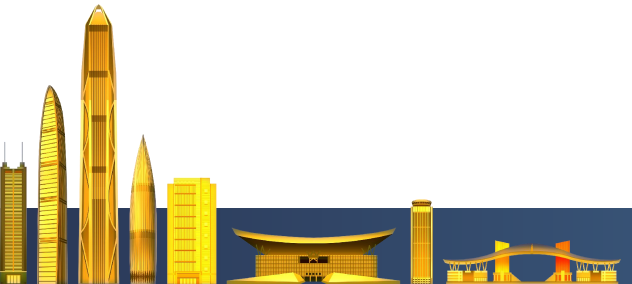
on a Raspberry Pi 5

Offline

questions never leave the device

Finding is done in advance; the model writes the answer.

Built with Bibliothèques Sans Frontières and the Dr. Denis Mukwege Foundation



People ask a health question in French, typos and all, and the phone answers in simple language, offline

User asks (in French, with spelling mistakes)

« quel sont les signe du paludisme »

Closest question found: Quels sont les symptômes courants du paludisme qu'il faut connaître ? (similarity 0.881)

Answer (written in advance, in simple French)

Les signes du paludisme qu'il faut connaître sont : • Avoir le corps chaud, c'est-à-dire avoir de la fièvre (39,5° ou plus) • Avoir des vomissements • … Si tu as un de ces signes, il faut aller vite voir un docteur ou aller au centre de santé.

(Malaria signs: fever of 39.5° or more, vomiting, … If you have one, see a doctor or go to the health centre quickly.)

80%

good answers on 200 test questions

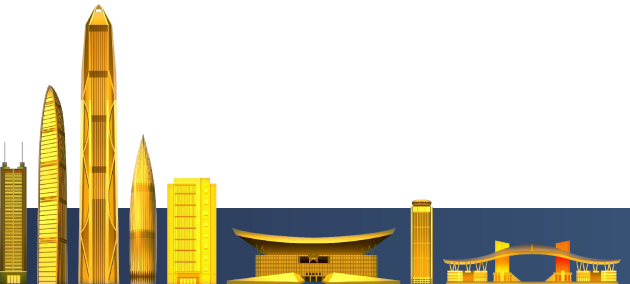
11,559

Q&A pairs, in French and Wolof

30 MB

search model, no network needed

Next: a relevance cutoff would cut bad answers shown from 20% to 11%



If the answer is written on the page, memory finds it. If it must be computed from the pages, memory doesn't.

1

Fixed documents

The corpus is known before the first question

2

Must run where the documents are

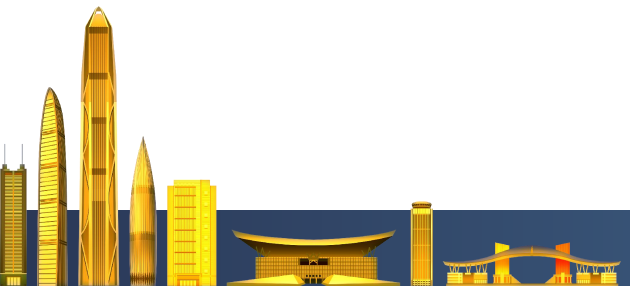
No cloud, no network: a phone, a Pi, a laptop

3

Cited answers

Every claim points to its page

A real office question can mean 400,000 tokens of spreadsheets. Nobody writes that answer in advance.



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03

When the answer must be computed



A few gigabytes must hold the model and its memory, and finding the right page is where small models fail.

Memory



Both share a few GB of RAM.

More text in context means a bigger KV cache and less room for the model.

Finding the right page

36% → **55%**

9B model — 27B model

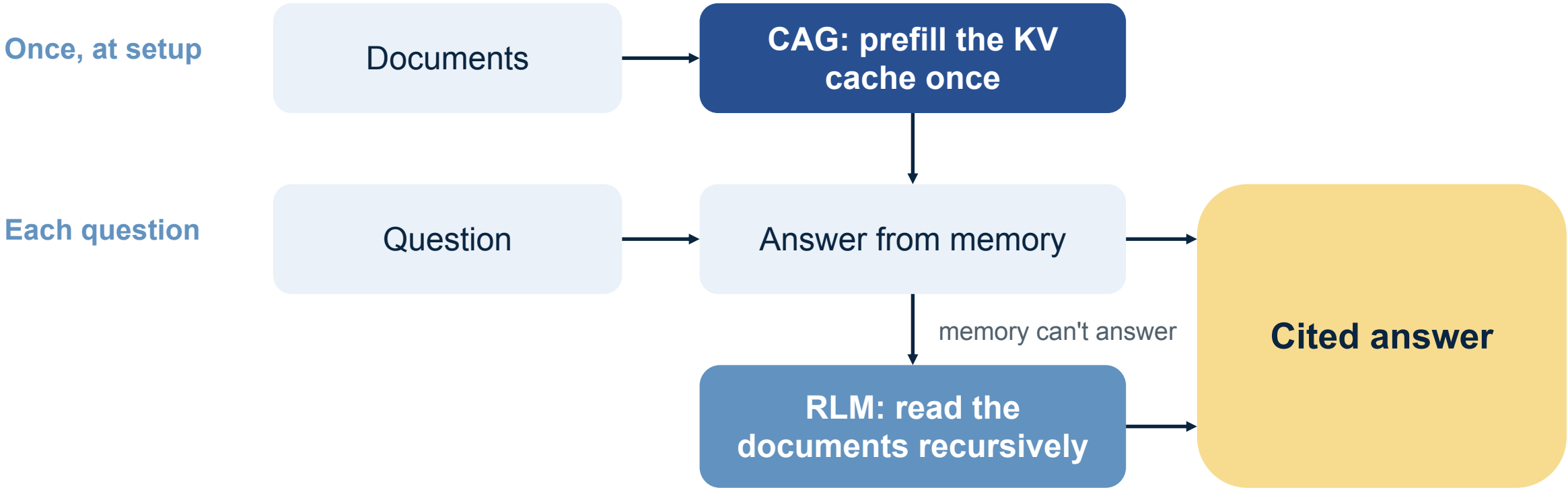
Search is where the small model loses most of its answers.

Measured on a server.

321M on the Pi (apps) · 4B / 9B / 27B measured on a server · 27B ≈ 21 GB at 4-bit, needs a 32 GB laptop

One pipeline: CAG + RLM

You never read a million tokens at question time: memory answers what is on the page, the loop computes what is not.



Same model, right file given: one call 3 of 133, the loop 43 (27B, server).

Memory arm measured as pages-in-prompt, not a KV prefill.

321M on the Pi (apps) · 4B / 9B / 27B measured on a server · 27B ≈ 21 GB at 4-bit, needs a 32 GB laptop

04

What we measured



What the loop buys on 133 office questions (OfficeQA Pro, server).

	one call, BM25 top-15 pages in prompt (~30k tokens)	loop, 14 turns
9B, full scope (2,500 pages)	0.8%	3.0%
27B, full scope	0.8%	15.8%
27B, right file given	2.3%	32.3%

When the right file is given, every correct loop answer followed a read of the right page; none without.

Exact match, one run at temperature 0, paired on the same questions, no LLM judge.

321M on the Pi (apps) · 4B / 9B / 27B measured on a server · 27B ≈ 21 GB at 4-bit, needs a 32 GB laptop

Four things that did not help: more turns, bigger prompts, guard rails, an LLM judge.

What we tried	What happened
More turns	9B, 14 → 40 turns, right file given: 24 → 24 correct (7 gained, 7 lost); 2× time, 2.5× tokens.
Bigger prompt	One call with the right page already in the prompt (110 of 133 questions): 0 correct at 9B, 1 at 27B.
Guard rails	Four enforcement mechanics fired 0, 0, 2, 0 times in 133 runs; the +10-point lift came from two sentences of prompt.
LLM judge	Checked against a human on 84 items: κ 0.61, below our 0.70 bar. Stronger judges agreed with the human less. GDPval scores are not reported.

321M on the Pi (apps) · 4B / 9B / 27B measured on a server · 27B ≈ 21 GB at 4-bit, needs a 32 GB laptop

The biggest lever: feedback text

After fixing only what the model was told about its errors, a 4B went from 0 to 4 solved tasks out of 5 (pilot, n = 1).

Before

```
Syntax error. Use: (grep "word")
malformed grep → []
```

0 / 5

tasks solved

After

```
parse error at …<here>… · typed
errors · "you sent the SAME code as
last turn"
```

4 / 5

tasks solved

Our Python loop, real tracebacks: 0 repeated turns in 91.
The DSL harness, canned "Unknown command": the same program four times in a row.

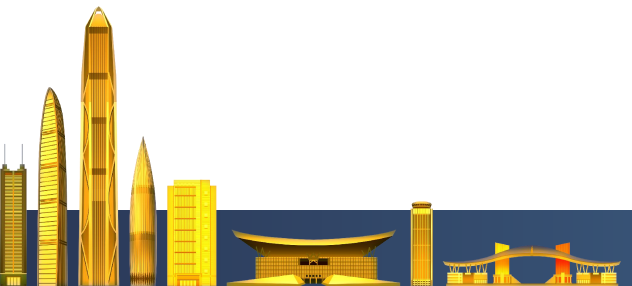
321M on the Pi (apps) · 4B / 9B / 27B measured on a server · 27B ≈ 21 GB at 4-bit, needs a 32 GB laptop

Takeaways

1. Small open models serve prepared, cited answers offline. Real knowledge work still needs model size.

2. You don't read the million tokens: prepare memory once (CAG), then read only what is missing (RLM).

3. Words beat machinery: two sentences of prompt moved the 27B more than any feature we added; better error text moved a 4B from 0 to 4 of 5.



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Thanks!

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